

40009 36

Haskell Calculus

Submitters

kw1425

18
—
20

Emarking

```
1: Final Tests: Summary for kw1425 of c1
2: PPT 21
3: -----
4:
5:   Public Tests:
6:     student-tests/calculus-test/calculus/sumTo:      2 / 2
7:     student-tests/calculus-test/calculus/vars:        7 / 7
8:     student-tests/calculus-test/calculus/inTermsOf:   8 / 8
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15:    original-tests/calculus-test/calculus/sumTo:      2 / 2
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21:    original-tests/calculus-test/calculus/maclaurin:    8 / 8
22:    original-tests/calculus-test/calculus/simplify expr: 1 / 1
23:    original-tests/calculus-test/calculus/pi:          3 / 3
24:
25: Git Repo: git@gitlab.doc.ic.ac.uk:lab2526_autumn/haskellcalculus_kw1425.git
26: Commit ID: dd555
27: Git Diff: https://gitlab.doc.ic.ac.uk/lab2526_autumn/haskellcalculus_kw1425/-/compare/master..dd555?from_project_id=132050
```

Final Tests	Tests.hs: 1/3	Katie Wan(kw1425):c1:21	Final Tests	Tests.hs: 2/3	Katie Wan(kw1425):c1:21
<pre> 1: {-# LANGUAGE StandaloneDeriving #-} 2: {-# OPTIONS_GHC -Wno-unrecognised-pragmas #-} 3: {-# HLINT ignore "Evaluate", ignore "Redundant negate" #-} 4: import Test.Tasty (defaultMain, TestTree, testGroup 5: , after, DependencyType(AllSucceed) 6: , adjustOption, mkTimeout, Timeout(..)) 7: import Test.Tasty.HUnit (testCase, Assertion) 8: import IC.Approx ((~>), (~>?)) 9: import IC.Exact ((-->)) 10: 11: import Prelude hiding (pi) 12: 13: import Expr 14: import Calculus 15: import Data.Map qualified as Map 16: import Data.Set qualified as Set 17: 18: main :: IO () 19: main = defaultMain tests 20: 21: tests :: TestTree 22: tests = testGroup "calculus" 23: [testGroup "sumTo" (numberedTests sumToTests) 24: , testGroup "vars" (numberedTests varsTests) 25: , after AllSucceed "/vars/" \$ 26: testGroup "inTermsOf" (numberedTests inTermsOfTests) 27: , testGroup "eval" (numberedTests evalTests) 28: , testGroup "class" (numberedTests classTests) 29: , after AllSucceed "/class/" \$ 30: testGroup "diff" (numberedTests diffTests) 31: , after AllSucceed "/diff/ /eval/" \$ 32: testGroup "maclaurin" (numberedTests maclaurinTests) 33: , after AllSucceed "/maclaurin/" \$ 34: testGroup "simplify expr" simplifyTests 35: , after AllSucceed "/class/ /eval/" \$ 36: testGroup "pi" (numberedTests piTests) 37:] 38: 39: sumToTests :: [Assertion] 40: sumToTests = [sumTo 4 [1..] --> 10 41: , sumTo 10 [1,3..] --> 100 42:] 43: 44: varsTests :: [Assertion] 45: varsTests = [vars e1 --> Set.fromList ["x"] 46: , vars e2 --> Set.fromList ["x", "y"] 47: , vars e3 --> Set.fromList ["x", "y"] 48: , vars e4 --> Set.fromList ["x"] 49: , vars e5 --> Set.fromList ["x"] 50: , vars e6 --> Set.fromList ["x"] 51: , vars (Val 5.0) --> Set.empty 52:] 53: 54: inTermsOfTests :: [Assertion] 55: inTermsOfTests = [inTermsOf "x" e1 --> True 56: , inTermsOf "x" e2 --> False 57: , inTermsOf "y" e3 --> False 58: , inTermsOf "x" e4 --> True 59: , inTermsOf "x" e5 --> True 60: , inTermsOf "y" e6 --> False 61: , inTermsOf "x" (Val 5.0) --> True 62: , inTermsOf "y" (Val 5.0) --> True 63:] 64: 65: evalTests :: [Assertion] 66: evalTests = [eval (Map.fromList [("x",380)]) (Val 7) --> 7.0 </pre>			<pre> 67: , eval (Map.fromList [("x",380), ("a",42), ("t",10)]) (Id "a") --> 7 68: , eval (Map.fromList [("t",10), ("t'",18)]) 69: (Bin Add (Val (-5)) (Id "t'")) --> 13.0 70: , eval (Map.fromList [("t",10), ("t'",19)]) 71: (Pre Neg (Bin Add (Val (-5)) (Id "t'"))) --> -14.0 72: , eval (Map.fromList [("t",10), ("t'",18.6), ("x",-55)]) 73: (Bin Mul (Id "x") (Id "x")) --> 3025.0 74: , eval (Map.fromList [("z",7)]) 75: (Bin Div (Val 3) (Id "z")) --> 0.42857142857142855 76: , eval (Map.fromList [("x",0.37)]) 77: (Pre Neg (Id "x")) --> -0.37 78: , eval Map.empty (Pre Sin (Val 2.4)) --> 0.675463180551151 79: , eval Map.empty (Pre Cos (Val 2.4)) --> -0.7373937155412454 80: , eval (Map.fromList [("x",0.37)]) e1 --> 1.85 81: , eval (Map.fromList [("x",0.37), ("y", 8.2)]) e2 --> 1.3369 82: , eval (Map.fromList [("x",0.37), ("y", 2.0)]) e3 --> 7 83: , eval (Map.fromList [("x",0.37)]) e4 --> (-0.9323273456060345) 84: , eval (Map.fromList [("x",0.37)]) e5 --> 0.6433720724587564 85: , eval (Map.fromList [("x",0.37)]) e6 --> 0.8799171617597958 86:] 87: 88: classTests :: [Assertion] 89: classTests = [(5 :: Expr) --> (5.0 :: Expr) 90: , Id "x" - 3 --> Id "x" + negate 3 91: , Id "x" ^ 3 --> Id "x" * Id "x" * Id "x" 92: , recip (Id "x") --> 1 / Id "x" 93: , Id "x" ^^ (-3) --> 1 / (Id "x" * Id "x" * Id "x") 94:] 95: 96: -- All of these tests have been written using the overloads, to ensure that 97: -- rewrite steps do not affect the testing 98: diffTests :: [Assertion] 99: diffTests = [diff "x" e1 --> 5 * 1 + 0 * Id "x" 100: , diff "x" e2 --> Id "x" * 1 + 1 * Id "x" + 0 - 0 101: , diff "y" e2 --> Id "x" * 0 + 0 * Id "x" + 1 - 0 102: , diff "x" e4 --> negate (negate (sin (Id "x") * 1)) 103: , diff "x" e5 --> cos (1 + log (2 * Id "x")) 104: , diff "x" e6 --> (3 * (Id "x" * 1 + 1 * Id "x")) 105: , diff "x" e6 --> (3 * (Id "x" * 1 + 1 * Id "x")) 106: , diff "x" e6 --> (3 * (Id "x" * 1 + 1 * Id "x")) 107: , diff "x" (exp (Id "x")) --> exp (Id "x") 108: , diff "x" (exp (Id "x" ^ 2)) --> 109: (Id "x" * 1 + 1 * Id "x") * exp (Id "x" ^ 2) 110:] 111: 112: 113: maclaurinTests :: [Assertion] 114: maclaurinTests = [maclaurin (Pre Sin (Id "x")) 2 2 -->? Just 2.0 115: , maclaurin (Pre Sin (Id "x")) 2 3 -->? Just 2.0 116: , maclaurin (Pre Sin (Id "x")) 2 5 -->? Just 0.6666666666666667 117: , maclaurin (Pre Sin (Id "x")) 2 7 -->? Just 0.9333333333333333 118: , maclaurin (Pre Sin (Id "x")) 2 9 -->? Just 0.9079365079365079 119: , maclaurin (Pre Cos (Id "x")) 4 9 -->? Just 7 120: , maclaurin (Pre Exp (Id "x")) 1 7 -->? Just (exp 1) 121: , maclaurin (Pre Sin (Id "y")) 1 5 -->? Nothing 122:] 123: 124: simplifyTests :: [TestTree] 125: simplifyTests = [withTimeout 100_000 \$ testCase "long series" 126: (maclaurin (Pre Sin (Id "x")) 2 30 -->? Just 0.909) 127:] 128: 129: piTests :: [Assertion] </pre>		
git@gitlab.doc.ic.ac.uk:lab2526_autumn/haskellcalculus_kw1425.git		dd555	git@gitlab.doc.ic.ac.uk:lab2526_autumn/haskellcalculus_kw1425.git		dd555

```
130: piTests = [ pi 1 --> 4
131:             , pi 10000 ~~> 3.142
132:             , eval Map.empty (cos (pi 25)) ~~> -1.0
133:             ]
134:
135: -----
136: -- HELPERS
137:
138: -- Why is this here? we _really_ want you to be using pattern matching at this
139: -- point.
140: deriving instance Eq Expr
141:
142: -- allows us to use 'pi :: Int -> Expr', for funsies.
143: instance Enum Expr where
144:   toEnum = Val . fromIntegral
145:   fromEnum ~(Val n) = floor n
146:
147: {-|
148: This function just matches up a bunch of assertions to a numerical naming
149: system, allowing us to distinguish them.
150:
151: If we wanted, we could provide descriptions to them instead...
152: -}
153: numberedTests :: [Assertion] -> [TestTree]
154: numberedTests = zipWith (\n -> testCase ("#" ++ show n)) ([1..] :: [Integer])
155:
156: withTimeout :: Integer -> TestTree -> TestTree
157: withTimeout n =
158:   adjustOption (\to -> if to == NoTimeout then mkTimeout n else to)
```

```

1: module Calculus (sumTo, vars, inTermsOf, eval, pretty, diff, maclaurin, pi) where
2:
3: import Prelude hiding (pi)
4:
5: import Expr
6:
7: import Data.List (scanl')
8:
9: import Data.Map (Map, (!))
10: import Data.Map qualified as Map
11:
12: import Data.Set (Set)
13: import Data.Set qualified as Set
14:
15: type Env = Map String Double
16:
17: -----
18: -- Type classes and class instances
19:
20: instance Num Expr where
21:   fromInteger :: Integer -> Expr
22:   fromInteger n = Val (fromInteger n)
23:
24:   negate      :: Expr -> Expr
25:   negate (Val 0) = Val 0
26:   negate x      = Pre Neg x
27:
28:   (+), (*)      :: Expr -> Expr -> Expr
29:   x + y = case (x, y) of
30:     (e, Val 0) -> e
31:     (Val 0, e) -> e
32:     _         -> Bin Add x y
33:
34:   x * y = case (x, y) of
35:     (Val 1, e) -> e
36:     (e, Val 1) -> e
37:     (Val 0, e) -> Val 0
38:     (e, Val 0) -> Val 0
39:     _         -> Bin Mul x y
40:
41: instance Fractional Expr where
42:   fromRational :: Rational -> Expr
43:   fromRational n = Val (fromRational n)
44:   (/)           :: Expr -> Expr -> Expr
45:   x / y = case (x, y) of
46:     (Val 0, e) -> 0
47:     (e, Val 1) -> e
48:     _         -> Bin Div x y
49:
50: instance Floating Expr where
51:   sin, cos, log, exp :: Expr -> Expr
52:   sin = Pre Sin
53:   cos = Pre Cos
54:   log = Pre Log
55:   exp = Pre Exp
56:
57: -----
58:
59: {-|
60: Computes the sum of first 'n' terms of the given series.
61: -}
62:
63: sumTo :: Num a => Int -> [a] -> a
64: -- sumTo = (sum .) . take -- nick told us not to use too much composition as it /
65: sumTo n = sum . take n

```

missing case for negate (Val n) = Val (negate n)

$$\frac{1+1.5}{3}$$

This was considered 'poor style'. It may be helpful if you need to maximise performance, but that is not the case here. readability is more important

is not very readable, but is this ok?

1/1

```

66:
67: {-|
68: Computes the set of variable names that appear within the given expression.
69: -}
70: vars :: Expr -> Set String
71: vars (Val _) = Set.empty
72: vars (Id name) = Set.singleton name
73: vars (Bin _ l r) = Set.union (vars l) (vars r)
74: vars (Pre _ r) = vars r
75:
76: {-|
77: Does the given expression only use the given variable and no others?
78: -}
79: inTermsOf :: String -> Expr -> Bool
80: inTermsOf x e = Set.size vs == 1
81:   where vs = Set.insert x (vars e)
82:
83: {-|
84: Evaluates a given expression, evaluating any variables to their value within
85: the provided environment.
86: -}
87: lookupBinOp :: Map BinOp (Double -> Double -> Double)
88: lookupBinOp = Map.fromList
89:   [ (Add, (+))
90:   , (Mul, (*))
91:   , (Div, (/))
92:   ]
93:
94: lookupPreOp :: Map PreOp (Double -> Double)
95: lookupPreOp = Map.fromList
96:   [ (Neg, negate)
97:   , (Sin, sin)
98:   , (Log, log)
99:   , (Cos, cos)
100:   , (Exp, exp)
101:   ]
102:
103: eval :: Env -> Expr -> Double
104: eval _ (Val x) = x
105: eval m (Id x) = m ! x
106: eval m (Bin op l r) = (lookupBinOp ! op) (eval m l) (eval m r)
107: eval m (Pre op r) = (lookupPreOp ! op) (eval m r)
108:
109: {-| OPTIONAL
110: Pretty prints an expression to a more human-readable form.
111: -}
112: class OpString a where
113:   opStr :: a -> String
114:
115: instance OpString BinOp where
116:   opStr Add = "+"
117:   opStr Mul = "*"
118:   opStr Div = "/"
119:
120: instance OpString PreOp where
121:   opStr Neg = "-"
122:   opStr Sin = "sin"
123:   opStr Log = "log"
124:   opStr Cos = "cos"
125:   opStr Exp = "exp"
126:
127: pretty :: Expr -> String
128: pretty (Val x) = show x
129: pretty (Id x) = x
130: pretty (Bin op l r) = "(" ++ pretty l ++ (opStr op) ++ pretty r ++ ")"
131: pretty (Pre op r) = (opStr op) ++ "(" ++ pretty r ++ ")"

```

2/2

0.5/1

→ explore using isSubsetOf, difference or delete.

avoid using '==' on set size

2/2

Final Tests	Calculus.hs: 3/4	Katie Wan(kw1425):c1:21	Final Tests	Calculus.hs: 4/4	Katie Wan(kw1425):c1:21
132:	133: {-		198:	199: -- maclaurin :: Expr -- ^ expression to approximate (with 'x' free)	
134: Symbolically differentiates a term with respect to a given identifier.			200: -- --> Double -- ^ value to give to 'x'		
135: -			201: -- --> Int -- ^ number of terms to expand		
136: diff :: String -> Expr -> Expr			202: -- --> Maybe Double -- ^ the approximate result		
137: diff _ (Val _) = 0			203: -- maclaurin e x n		
138: diff x (Id y)		3/3	204: -- n <= 0 not (inTermsOf "x" e) = Nothing		
139: x == y = 1			205: -- otherwise = Just approx		
140: otherwise = 0			206: -- where		
141: diff x (Bin Add e1 e2) = (diff x e1) + (diff x e2)			207: -- z = Map.fromList [("x", 0)]		
142: diff x (Bin Mul e1 e2) = e1 * (diff x e2) + (diff x e1) * e2			208: -- es = iterate (diff "x") e -- [f(x), f'(x), f''(x)..]		
143: diff x (Bin Div e1 e2) = (e2 * (diff x e1) - e1 * (diff x e2)) / (e2 * e2)			209: -- f0 = map (eval z) es -- [f(0), f'(0), f''(0)..]		
144: diff x (Pre Sin e) = cos e * (diff x e)			210: -- facts = scanl' (*) 1 [1..] -- [1, 1!, 2!, 3!..]		
145: diff x (Pre Cos e) = negate ((sin e) * (diff x e))			211: -- coeffs = f0 / facts -- [a0, a1, a2, a3..]		
146: diff x (Pre Log e) = (diff x e) / e			212: -- xs = (repeat x) ** [0..] -- [1, x, x^2, x^3..]		
147: diff x (Pre Exp e) = (diff x e) * (Pre Exp e)			213: -- approx = sumTo n (coeffs * xs)		
148: diff x (Pre Neg e) = negate (diff x e)			214:		
149:					
150: {-					
151: Computes the approximation of an expression 'f' by expanding the Maclaurin					
152: series on 'f' and taking its summation.					
153: -					
154: -- original					
155: maclaurin :: Expr -- ^ expression to approximate (with 'x' free)					
156: --> Double -- ^ value to give to 'x'					
157: --> Int -- ^ number of terms to expand					
158: --> Maybe Double -- ^ the approximate result		4+0 s			
159: maclaurin e x n					
160: n <= 0 not (inTermsOf "x" e) = Nothing					
161: otherwise = Just approx					
162: where					
163: z = Map.fromList [("x", 0)]					
164: es = iterate (diff "x") e -- [f(x), f'(x), f''(x)..]					
165: f0 = map (eval z) es -- [f(0), f'(0), f''(0)..]		→ good comments			
166: facts = scanl' (*) 1 [1..] -- [1, 1!, 2!, 3!..]					
167: coeffs = zipWith (/) f0 facts -- [a0, a1, a2, a3..]					
168: xs = iterate (*x) 1 -- [1, x, x^2, x^3..]					
169: approx = sumTo n (zipWith (*) coeffs xs)					
170:					
171:					
172: -- Extension					
173: instance Num a => Num [a] where					
174: fromInteger :: Integer -> [a]					
175: fromInteger n = repeat (fromInteger n)					
176: negate :: [a] -> [a]					
177: negate = map negate		1.5/1.5			
178: (+), (*) :: [a] -> [a] -> [a]					
179: (+) = zipWith (+)					
180: (*) = zipWith (*)					
181:					
182: instance Fractional a => Fractional [a] where					
183: fromRational :: Rational -> [a]					
184: fromRational n = repeat (fromRational n)					
185: (/) :: [a] -> [a] -> [a]					
186: (/) = zipWith (/)					
187:					
188: instance Floating a => Floating [a] where					
189: (**) :: [a] -> [a] -> [a]					
190: (**) = zipWith (**)					
191:					
192: pi :: (Enum a, Fractional a) => Int -> a					
193: pi n = 4 * (sumTo n terms)					
194: where	✓	1.5 1.5			
195: num = cycle [1, -1]					
196: den = [1, 3..]					
197: terms = num / den					

Final Tests

testResults.txt: 1/5

Katie Wan(kw1425):c1:21

```
1: ----- Test Output -----
2: copying calculus.cabal from skeleton
3: copying cabal.project from skeleton
4: Config file path source is default config file.
5: Config file not found: /root/.config/cabal/config
6: Writing default configuration to /root/.config/cabal/config
7: Resolving dependencies...
8: Build profile: -w ghc-9.6.7 -O1
9: In order, the following will be built (use -v for more details):
10: - calculus-0.1.0.0 (lib) (first run)
11: - call-stack-0.4.0 (lib) (requires build)
12: - colour-2.3.6 (lib) (requires build)
13: - prettyprinter-1.7.1 (lib) (requires build)
14: - tagged-0.8.9 (lib) (requires build)
15: - ansi-terminal-types-1.1.3 (lib) (requires build)
16: - ansi-terminal-1.1.3 (lib) (requires build)
17: - prettyprinter-ansi-terminal-1.1.3 (lib) (requires build)
18: - optparse-applicative-0.19.0.0 (lib) (requires build)
19: - tasty-1.5.3 (lib) (requires build)
20: - tasty-hunit-0.10.2 (lib) (requires build)
21: - calculus-0.1.0.0 (test:calculus-test) (first run)
22: Configuring library for calculus-0.1.0.0...
23: Starting colour-2.3.6 (lib)
24: Starting call-stack-0.4.0 (lib)
25: Starting prettyprinter-1.7.1 (lib)
26: Building colour-2.3.6 (lib)
27: Building call-stack-0.4.0 (lib)
28: Building prettyprinter-1.7.1 (lib)
29: Preprocessing library for calculus-0.1.0.0...
30: Building library for calculus-0.1.0.0...
31: [1 of 2] Compiling Expr      ( src/Expr.hs, /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/build/Expr.o, /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/build/Expr.dyn_o )
32: [2 of 2] Compiling Calculus ( src/Calculus.hs, /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/build/Calculus.o, /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/build/Calculus.dyn_o )
33: Installing call-stack-0.4.0 (lib)
34: Completed call-stack-0.4.0 (lib)
35: Starting tagged-0.8.9 (lib)
36: Building tagged-0.8.9 (lib)
37: Installing tagged-0.8.9 (lib)
38: Installing colour-2.3.6 (lib)
39: Completed tagged-0.8.9 (lib)
40: Completed colour-2.3.6 (lib)
41: Starting ansi-terminal-types-1.1.3 (lib)
42: Building ansi-terminal-types-1.1.3 (lib)
43: Installing ansi-terminal-types-1.1.3 (lib)
44: Completed ansi-terminal-types-1.1.3 (lib)
45: Starting ansi-terminal-1.1.3 (lib)
46: Building ansi-terminal-1.1.3 (lib)
47: Installing prettyprinter-1.7.1 (lib)
48: Completed prettyprinter-1.7.1 (lib)
49: Installing ansi-terminal-1.1.3 (lib)
50: Completed ansi-terminal-1.1.3 (lib)
51: Starting prettyprinter-ansi-terminal-1.1.3 (lib)
52: Building prettyprinter-ansi-terminal-1.1.3 (lib)
53: Installing prettyprinter-ansi-terminal-1.1.3 (lib)
54: Completed prettyprinter-ansi-terminal-1.1.3 (lib)
55: Starting optparse-applicative-0.19.0.0 (lib)
56: Building optparse-applicative-0.19.0.0 (lib)
57: Installing optparse-applicative-0.19.0.0 (lib)
58: Completed optparse-applicative-0.19.0.0 (lib)
59: Starting tasty-1.5.3 (lib)
```

Final Tests

testResults.txt: 2/5

Katie Wan(kw1425):c1:21

```
60: Building      tasty-1.5.3 (lib)
61: Installing    tasty-1.5.3 (lib)
62: Completed     tasty-1.5.3 (lib)
63: Starting      tasty-hunit-0.10.2 (lib)
64: Building      tasty-hunit-0.10.2 (lib)
65: Installing    tasty-hunit-0.10.2 (lib)
66: Completed     tasty-hunit-0.10.2 (lib)
67: Configuring test suite 'calculus-test' for calculus-0.1.0.0...
68: Preprocessing test suite 'calculus-test' for calculus-0.1.0.0...
69: Building test suite 'calculus-test' for calculus-0.1.0.0...
70: [1 of 3] Compiling IC.Approx      ( test/IC/Approx.hs, /
/tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/build/calculus-test/calculus-test-tmp/IC/Approx.o )
71: [2 of 3] Compiling IC.Exact      ( test/IC/Exact.hs, /
/tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/build/calculus-test/calculus-test-tmp/IC/Exact.o )
72: [3 of 3] Compiling Main          ( test/Tests.hs, /
/tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/build/calculus-test/calculus-test-tmp/Main.o )
73: [4 of 4] Linking /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/build/calculus-test/calculus-test
74: Build profile: -w ghc-9.6.7 -O1
75: In order, the following will be built (use -v for more details):
76: - calculus-0.1.0.0 (test:calculus-test) (ephemeral targets)
77: Preprocessing test suite 'calculus-test' for calculus-0.1.0.0...
78: Building test suite 'calculus-test' for calculus-0.1.0.0...
79: Running 1 test suites...
80: Test suite calculus-test: RUNNING...
81: calculus
82:   sumTo
83:     #1:          OK
84:     #2:          OK
85:   vars
86:     #1:          OK
87:     #2:          OK
88:     #3:          OK
89:     #4:          OK
90:     #5:          OK
91:     #6:          OK
92:     #7:          OK
93:   inTermsOf
94:     #1:          OK
95:     #2:          OK
96:     #3:          OK
97:     #4:          OK
98:     #5:          OK
99:     #6:          OK
100:    #7:          OK
101:    #8:          OK
102:   eval
103:     #1:          OK
104:     #2:          OK
105:     #3:          OK
106:     #4:          OK
107:     #5:          OK
108:     #6:          OK
109:     #7:          OK
110:     #8:          OK
111:     #9:          OK
112:    #10:          OK
113:    #11:          OK
114:    #12:          OK
115:    #13:          OK
116:    #14:          OK
117:    #15:          OK
```


Final Tests**testResults.txt: 3/5****Katie Wan(kw1425):c1:21**

```
118: class
119:   #1:      OK
120:   #2:      OK
121:   #3:      OK
122:   #4:      OK
123:   #5:      OK
124: diff
125:   #1:      OK
126:   #2:      OK
127:   #3:      OK
128:   #4:      OK
129:   #5:      OK
130:   #6:      OK
131:   #7:      OK
132:   #8:      OK
133: maclaurin
134:   #1:      OK
135:   #2:      OK
136:   #3:      OK
137:   #4:      OK
138:   #5:      OK
139:   #6:      OK
140:   #7:      OK
141:   #8:      OK
142: simplify expr
143:   long series: OK
144: pi
145:   #1:      OK
146:   #2:      OK
147:   #3:      OK
148:
149: All 57 tests passed (0.00s)
150:
151: Test suite calculus-test: PASS
152: Test suite logged to:
153: /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/test/calculus-0.1.0.0-calculus-test.log
154: 1 of 1 test suites (1 of 1 test cases) passed.
155: copying test from skeleton
156: Build profile: -w ghc-9.6.7 -O1
157: In order, the following will be built (use -v for more details):
158: - calculus-0.1.0.0 (test:calculus-test) (file test/Tests.hs changed)
159: Preprocessing test suite 'calculus-test' for calculus-0.1.0.0...
160: Building test suite 'calculus-test' for calculus-0.1.0.0...
161: [3 of 3] Compiling Main                ( test/Tests.hs, /
/tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/build/calculus-test/calculus-test-tmp/Main.o ) [Source file changed]
162: [4 of 4] Linking /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/build/calculus-test/calculus-test [Objects changed]
163: Build profile: -w ghc-9.6.7 -O1
164: In order, the following will be built (use -v for more details):
165: - calculus-0.1.0.0 (test:calculus-test) (ephemeral targets)
166: Preprocessing test suite 'calculus-test' for calculus-0.1.0.0...
167: Building test suite 'calculus-test' for calculus-0.1.0.0...
168: Running 1 test suites...
169: Test suite calculus-test: RUNNING...
170: calculus
171:   sumTo
172:     #1:      OK
173:     #2:      OK
174:   vars
175:     #1:      OK
176:     #2:      OK
177:     #3:      OK
```

Final Tests**testResults.txt: 4/5****Katie Wan(kw1425):c1:21**

```
178:      #4:      OK
179:      #5:      OK
180:      #6:      OK
181:      #7:      OK
182: inTermsOf
183:      #1:      OK
184:      #2:      OK
185:      #3:      OK
186:      #4:      OK
187:      #5:      OK
188:      #6:      OK
189:      #7:      OK
190:      #8:      OK
191: eval
192:      #1:      OK
193:      #2:      OK
194:      #3:      OK
195:      #4:      OK
196:      #5:      OK
197:      #6:      OK
198:      #7:      OK
199:      #8:      OK
200:      #9:      OK
201:     #10:      OK
202:     #11:      OK
203:     #12:      OK
204:     #13:      OK
205:     #14:      OK
206:     #15:      OK
207: class
208:      #1:      OK
209:      #2:      OK
210:      #3:      OK
211:      #4:      OK
212:      #5:      OK
213: diff
214:      #1:      OK
215:      #2:      OK
216:      #3:      OK
217:      #4:      OK
218:      #5:      OK
219:      #6:      OK
220:      #7:      OK
221:      #8:      OK
222: maclaurin
223:      #1:      OK
224:      #2:      OK
225:      #3:      OK
226:      #4:      OK
227:      #5:      OK
228:      #6:      OK
229:      #7:      OK
230:      #8:      OK
231: simplify expr
232:   long series: OK
233: pi
234:      #1:      OK
235:      #2:      OK
236:      #3:      OK
```

```
237:
238: All 57 tests passed (0.00s)
```

Final Tests

testResults.txt: 5/5

Katie Wan(kw1425):c1:21

```
239:
240: Test suite calculus-test: PASS
241: Test suite logged to:
242: /tmp/d20251031-34-ntlr21/dist-newstyle/build/x86_64-linux/ghc-9.6.7/calculus-0.1.0.0/t/calculus-test/test/calculus-0.1.0.0-calculus-test.log
243: 1 of 1 test suites (1 of 1 test cases) passed.
244:
245: ----- Test Errors -----
246: Warning: The package list for 'hackage.haskell.org' is 38 days old.
247: Run 'cabal update' to get the latest list of available packages.
```